ANPRE: An End-to-End Neuroinformatics Toolkit for Digitizing Clinical Neuropsychological Data

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Introduction

Current Landscape

- Cataloguing of **neuropsycho-logical assessment** performance is analog and clinic-specific
- **Little documentation** exists on the best practices for extracting score data
- Clinicians are **not commonly trained** on data automation technology

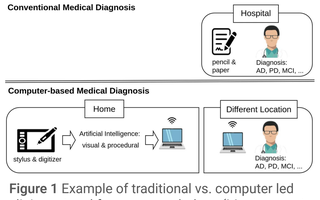


Figure 1 Example of traditional vs. computer led clinic protocol for neuropsychology (Moetesum et al., 2022)

- Biomedical informatics, which has undergone rapid horizontal research development[3], has established the foundation for how to implement data automation in clinical neuropsychology[2]
- Inherent difficulties in managing sensitive patient data while maintaining privacy protections[1], [4]

Research Aims

- Develop a **computer vision-based neuroinformatic pipeline** to process neuropsychological assessments at scale.
- Maintain **toolkit organization** to allow both inference-level and development-level adaptation as a proof-of-concept

Data and Methods

The Benton Neuropsychology Clinic's datasheets (BNCSS) contains most quantitative data from the battery for evaluating a patient.

- **Patient Information Block:** such as name, DOB, referral information, handedness, education
  - ➢ De-identification would usually remove this section
- **Field Labels:** each row, representing tests (e.g. `WAIS-IV`), test categories (`MEMORY`), or subtests (`Rey-Osterrieth: Copy`)
- **Data Groups:** each of 3 main columns, resp.
  - ➢ **Test Information:** test parameters, patient time taken, errors made
  - ➢ **Raw Score**
  - ➢ **Clinician Comments:** percentiles, confidence intervals, norms

Constraints of Input:

+ BNCSS can be found within the **first ~10 pages** of case files → faster runtimes
+ Patient info is **in metadata,** accessible without processing

- Scan sheet **need not be straight**
- Handwriting quality can be **poor**
- Handwriting can lie **outside** or **between** the the frame

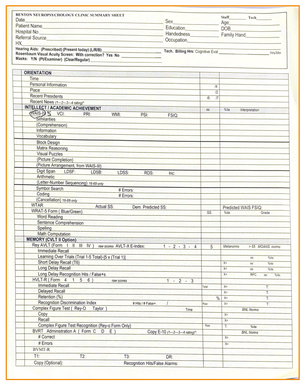
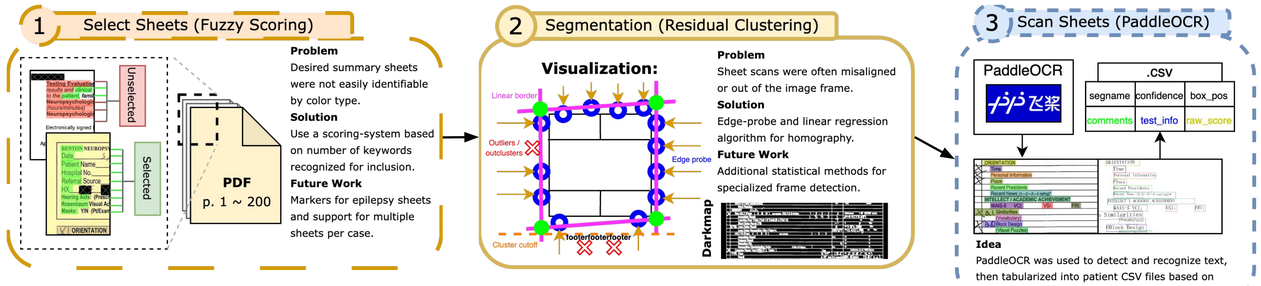


Figure 2 Empty BNCSS sample. As seen, the scan is misaligned (rotated and skewed), which may affect text quality. This type of error is corrected for in 2. Segmentation.

ANPRE Multi-Modular Framework Development



Patient Processing Pipeline

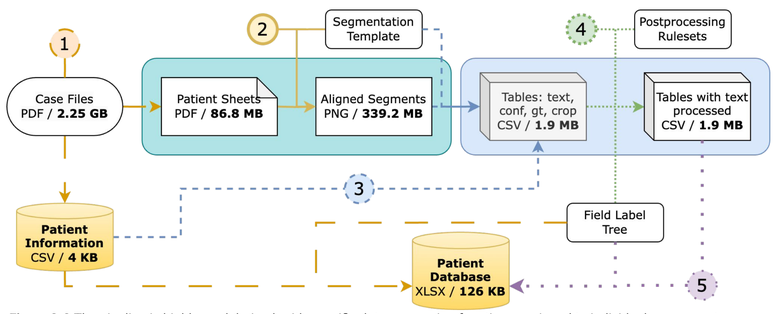
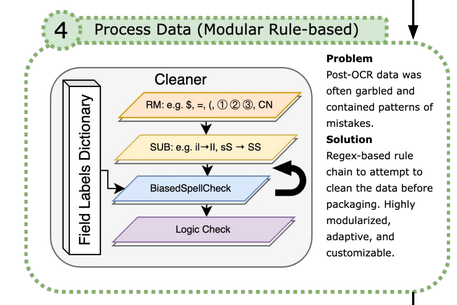


Figure 3-8 The pipeline is highly modularized, with specific data processing functions assigned to individual components.



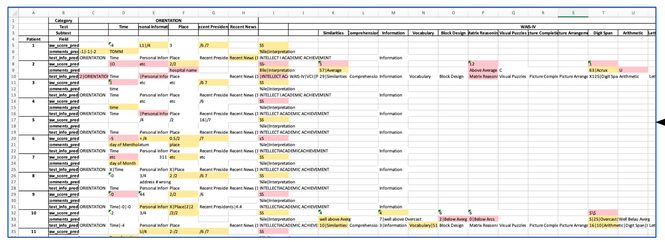
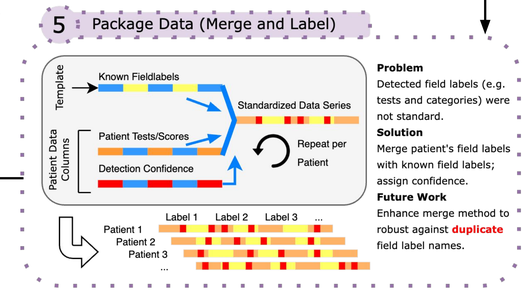


Figure 9 Instance of patient data spreadsheet. Rows: patients. Subrows: data groups. Columns: categories, tests, and subtests. Red and yellow indicates low-confidence. Data was jumbled for demonstration.



Results → Quantitative Performance Analysis

Engineering Experiment Design

- ➢ **Dataset Overview:** Random sample of `n=30` from patient cases conducted in 2025
- ➢ **Implementation:**
  - **Computer Cases**: Implement in Python (~2200 lines)
  - **Human Cases:** Token crops (groundtruth) lined up along detect text, holistically scored by me as Wrong, Recoverable, or Correct for data capture acc. (DCA).
- ➢ **Explanatory data groups:**
  - **Field Labels**: either all field labels or only Orientation+WAIS were included
  - **Handwriting**: Handwriting (HW) and no-HW groups were measured separately. No-HW usually indicates no data / the test wasn't performed

Table 1 Comparison of performance times for manual data entry, automated entry with Computer A (Macbook CPU), and data entry with Computer B (Windows GPU)

Pipeline	Average Time per Patient (s)				
	Human	Computer A (CPU)		Computer B (GPU)	
Selection	~30	0.705	1.90%	~0.705	~
Segmentation		0.506	1.40%	~0.506	~
Text Detection	~420	34.796	96.20%	2.524	60.64%
Processing	~180	0.361	1.00%	0.392	9.42%
Synthesis		0.037	0.10%	0.035	0.84%
Time / patient	**10.5 min**		**36.405 s**		**4.162 s**

Score Capture Performance

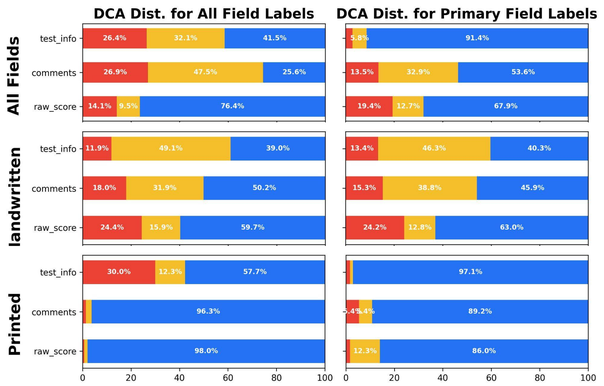


Figure 10 The left column represents data capture accuracy (DCA) distribution for all field labels (categories, tests, subtests), while the right column represents DCA for only Orientation and WAIS (Intellect), which can be applied the most generally. The first row represents both HW and no-HW data; the second row includes with-HW data; and the third row includes only no-HW data.

Results → Metrics

For 100 patients...

- (S1) Accuracy of selecting BNCSS: **99%**
- (S2) Accuracy of segmenting BNCSS: **97.25%**
- If accurate, potential improvements include
  - ➢ Reduction in data collection time: **10 minutes → 4 seconds ∴ 99.333% time** and **work** saved
  - ➢ Reduction in amount of data to be analyzed. **2.25 GB → 141 KB ∴ 99.997% processing data** saved

Sheet Selection Performance

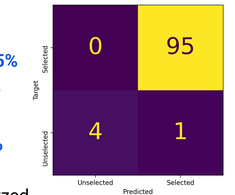


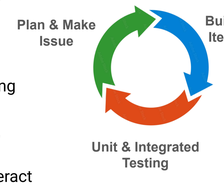
Figure 11 Confusion matrix of accuracy of sheet selection. One epilepsy sheet was incorrectly selected, making a false positive.

Discussion

- Convert raw patient casefiles → **standardized** .xlsx database in format prepared for analysis and research
- **Confidence-level** labeling for clinical feasibility
- The ANPRE toolkit serves as a **starting point** for digitizing neuropsychological data with minimal interference to clinical procedure
- Multimodular framework enables continued development and accuracy optimization for ANPRE

Skill Development

- Designed algorithms **from scratch**, using concepts from well-known techniques: fuzzy-logic, coordinate clustering, regex, rulesets, feature-extraction
- Built a software and programming workflow
  - ➢ Developed a **multi-module** project with interacting components, which helped significantly with debugging and reworking previous code
  - ➢ Improved Github usage autonomy: trees, pre-commits, etc.
- Interacted with toolkits: PaddleOCR, PyTesseract



Future Directions and Continued Pipeline Development

- ➔ Develop textual groundtruth dataset to increase sample size and facilitate future development and testing
- ➔ Develop a **data labelling GUI** to adapt new tests, templates, and field label corrections for broader clinical applications
- ➔ Detect single data cells for strictly-aligned text detection
- ➔ Explore alternatives to PaddleOCR, with consideration of data confidentiality in mind
- ➔ Conduct integration testing and a lesion-mapping case using this data

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